



CIE: Continuous Internal Evaluation
SEE: Semester End Examination

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Day	8:30-10:00	10:00-11:30	11:30-1:00	1:00-2:30	2:30-4:00	4:00-5:30
Saturday						
Sunday						
Monday						
Tuesday						
Wednesday						
Thursday	Day Off					
Friday						

Course Content (from syllabus)

Foundation of Digital Electronics, Fundamental concepts, digital and analog quantities, advantages of digital electronics, Introduction to Number Systems, Conversion Number Systems, Expression, truth table, universal properties of NAND & NOR gates, Simplification of expression using Boolean algebra, Binary Subtraction using 1's and 2's complements, Design of Combinational Logic using TT, Boolean algebra, Adder Circuit, Half Adder, Full Adder, Parallel Adder, Introduction to K-Map, SOP, POS, MINTERM, MAXTERM, 3-4 variable K- MAP, construction of map, implementation of expression using map, grouping and simplification, Advanced Combinational Circuits, Decoder (Binary to Decimal, Octal, Hexadecimal), Advanced Combinational Circuits, Converters (BCD to XS-3), Introduction to Multiplexers and De-multiplexer, Design of Multiplexer Circuits, Design of De-multiplexer Circuits, Introduction to Parity, Design of Parity Generator Circuits, Design of Parity Checker Circuit to detect error during transfer or transmission of Data, Introduction to comparator circuit using complex combinational Logic, Introduction to Computer Memory, Introduction to Flip Flops, Design of J-K, S-R, Latch.

Rationale of the Course

The provided content outlines the foundational concepts covered in a course on Digital Electronics. It begins with an introduction to fundamental concepts, encompassing both digital and analog quantities, and highlights the advantages of digital electronics. The curriculum progresses to cover Number Systems, including conversion techniques and expressions, truth tables, and the universal properties of NAND and NOR gates. It delves into the simplification of expressions using Boolean algebra and explores binary subtraction through 1's and 2's complements. The design of combinational logic is introduced, utilizing truth tables, Boolean algebra, and various adder circuits such as Half Adder, Full Adder, and Parallel Adder. The course further advances to cover topics like Karnaugh Maps (K-Maps), SOP and POS expressions, MINTERMs, MAXTERMs, and their construction and implementation. It explores advanced combinational circuits, including decoders, converters (BCD to XS-3), and the introduction to multiplexers and demultiplexers. The content also includes the design of circuits related to parity, addressing error detection during data transfer. The latter part introduces the concept of a comparator circuit using complex combinational logic, explores computer memory, and delves into flip-flops, specifically J-K, S-R, and latches. Overall, the content comprehensively covers essential digital electronics topics, providing a solid foundation for understanding and designing various digital circuits and systems.

Course Objectives

The objective of this course is to deliver insights into digital electronics and logical designs as the circuits in the real-world works, specifically:

- To learn to represent numerical values in various number systems,
- To learn to demonstrate knowledge of logic gates, Boolean algebraic manipulation
- To learn to apply theorems and mappings for simplification of expressions

- To learn to design and implement types of decoders, encoders, multiplexers, and De-multiplexers.
- To learn to apply circuits for developing flip-flops and memory registers

Course Learning Outcomes (CLO) with Mappings

At the end of the course, students will be able to:

CLO's	CLO Descriptions	Program Learning Outcomes (PLOs)	Learning Domains (C, P, A)	BNQF Skill
CLO1	Estimate the concept of representation of numerical values in various number systems.	PLO2	C2	Fundamental Domain
CLO2	Apply the knowledge of logic gates, Boolean algebra for implementation and manipulation of various logic circuits using simplification techniques i.e. K-map.	PLO1	C3	Fundamental Domain
CLO3	Design combinational circuits for decoders, encoders, multiplexers, de-multiplexers and others complex circuits.	PLO3	C3	Fundamental Domain
CLO4	Construct sequential circuits for flip-flops, registers, and memory of computing devices.	PLO3	C3	Fundamental Domain

Mapping of CLOs with PLOs

Course Learning Outcome	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2	√											
CLO3			√									
CLO4			√									

Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week/ Lesson (hour)	Lesson Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
Week-1 Lesson 1 & 2 [3 Hours]	Lesson 1: Foundation of Digital Electronics, Fundamental concepts, digital and analog quantities, advantages of digital electronics.	Classroom discussion, Open discussion.	Class Test, Midterm exam	CLO1

	Lesson 2: Introduction to Number Systems, Conversion between different Number Systems	Classroom discussion, Voice over PPT, Lecture video, Lecture notes, Open discussion.	Class Test, Midterm exam	CLO1
Week-2 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Digital Logic & Gates	Brainstorming sessions, Lecture video, Lecture notes,	Class Test, Midterm exam	CLO1
	Lesson 2: Expression, truth table, universal properties of NAND & NOR gates	Brainstorming sessions, Lecture video, Lecture notes, Open discussion.	Class Test, Midterm exam	CLO1
Week-3 Lesson 1 & 2 [3 Hours]	Lesson 1: Boolean Algebra, Theorems and postulates, proof of theorem.	Classroom discussion, Voice over PPT, Lecture video, Lecture notes	Class Test, Assignment, Midterm exam	CLO2
	Lesson 2: Simplification of expression using Boolean algebra.	Classroom discussion, Voice over PPT, Lecture video, Lecture notes	Class Test, Assignment, Midterm exam	CLO2
Week-4 Lesson 1 & 2 [3 Hours]	Lesson 1: Binary Addition.	Classroom discussion, Voice over PPT, Lecture video, Lecture notes.	Class Test, Assignment, Midterm exam	CLO1
	Lesson 2: Binary Subtraction using 1's and 2's complements. Class Test -1	Classroom discussions, Voice over PPT, Lecture video, Lecture note.	Class Test, Assignment, Midterm exam	CLO1
Week-5 Lesson 1 & 2 [3 Hours]	Lesson 1: Design of Combinational Logic using TT, Boolean algebra.	Classroom discussions, Voice over PPT, Lecture video, Lecture note	Class Test, Assignment, Midterm exam	CLO3
	Lesson 2: Adder Circuit, Half Adder, Full Adder, Parallel Adder.	Classroom discussions, Voice over PPT, Lecture video, Lecture note	Class Test, Assignment, Midterm exam	CLO3
Week-6 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to K-Map, SOP, POS, MINTERM, MAXTERM.	Classroom discussion, Voice over PPT, Lecture video, Lecture notes, Open discussion.	Class Test, Assignment, Midterm exam	CLO2

	Lesson 2: 3-4 variable K- MAP, construction of map, implementation of expression using map, grouping and simplification Class Test -2	Classroom discussion, Lecture notes, Open discussion	Class Test, Assignment, Midterm exam	CLO2
Week-7 Lesson 1 & 2 [3 Hours]	Lesson 1: Advanced Combinational Circuits, Encoder (Decimal, Octal, Hexadecimal to Binary).	Classroom discussions, Voice over PPT, Lecture video, Lecture note, Open discussion.	Class Test, Assignment, Final exam	CLO3
	Lesson 2: Advanced Combinational Circuits, Decoder (Binary to Decimal, Octal, Hexadecimal)	Classroom discussions, Voice over PPT, Lecture video, Lecture note, Open discussion.	Class Test, Assignment, Final exam	CLO3
Week-8 Lesson 1 & 2 [3 Hours]	Lesson 1: Advanced Combinational Circuits, Converters (BCD to XS-3)	Classroom discussions, Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Assignment, Final exam	CLO3
	Lesson 2: Introduction to Multiplexers and De-multiplexer.	Brainstorming sessions, Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Assignment, Final exam	CLO3
Week-9 Lesson 1 & 2 [3 Hours]	Lesson 1: Design of Multiplexer Circuits,	Brainstorming sessions, Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Assignment, Final exam	CLO3
	Lesson 2: Design of De-multiplexer Circuits.	Brainstorming sessions, Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Assignment, Final exam	CLO3
Week-10 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Parity, Design of Parity Generator Circuits.	Voice over PPT, Lecture video, Lecture note, Open discussion	Class Test, Assignment, Final exam	CLO3
	Lesson 2: Design of Parity Checker Circuit to detect error during transfer or	Classroom discussions, Voice over PPT, Lecture	Class Test, Assignment, Final exam	CLO3

	transmission of Data.	video, Lecture note, Open discussion		
Week-11 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to comparator circuit using complex combinational Logic. Class Test -3	Classroom discussions, Voice over PPT, Lecture video, Lecture note, Open discussion	Assignment, Final exam	CLO3
	Lesson 2: Multiplexer circuit for representation of Boolean expression.	Classroom discussions, Voice over PPT, Lecture video, Lecture note, Open discussion	Assignment, Final exam	CLO3
Week-12 Lesson 1 & 2 [3 Hours]	Lesson 1: Introduction to Sequential Circuit, Introduction to Latches & Flip Flops	Classroom discussions, Voice over PPT, Lecture video, Lecture note, Open discussion	Assignment, Final exam	CLO4
	Lesson 2: Design of J-K, S-R, D, T Flip flop.	Classroom discussions, Voice over PPT, Lecture video, Lecture note, Open discussion	Assignment, Final exam	CLO4
Week-13 Lesson 1 & 2 [3 Hours]	Lesson 1: Basics of Register, Designing of various types of registers i.e. SISO, SIPO, PISO, PIPO etc.	Classroom discussions, Voice over PPT, Lecture video, Lecture note, Open discussion	Assignment, Final exam	CLO4
	Lesson 2: Introduction to Computer Memory, Designing of PLA, RAM etc.	Classroom discussions, Voice over PPT, Lecture video, Lecture note, Open discussion	Assignment, Final exam	CLO4
Week-14 Lesson 1 & 2 [3 Hours]	Lesson 1: Industry Session	Interactive Lectures and Presentations, Expert Talks, Hands- On Workshops, Site Visits and Virtual Tours	Assignment, Presentation	CLO1, CLO2, CLO3, CLO4
	Lesson 2: Review class on final syllabus topic	Brainstorming sessions, Open discussion	None	CLO2, CLO3, CLO4

Overall Assessment Scheme

Assessment Task	CLO's				Mark (Total=100)	PLO's			Mark (Total =68)
	CLO1	CLO2	CLO3	CLO4		PLO2	PLO1	PLO3	
Attendance	--	--	--		7				
Class Test 1	5(15)				Avg 15				
Class Test 2		5(15)							
Class Test 3			5(15)						
Assignment		5			5				
Presentation		4	4		8				
Midterm Exam	10	15			25	10	15		25
Final exam		10	15	15	40		10	30	40

Marks Distribution

Class attendance	7
Assignment	5
Presentation (Mandatory)	8
3 Quizzes	15
Midterm Test	25
Semester Final Examination	40
Total	100

Evaluation Policy (Grading Policy)

Marks obtained out of 100	Grade	Grade point equivalent	Remarks
80% and above	A+	4.00	Outstanding
75% to less than 80%	A	3.75	Excellent
70% to less than 75%	A-	3.50	Very Good
65% to less than 70%	B+	3.25	Good
60% to less than 65%	B	3.00	Satisfactory
55% to less than 60%	B-	2.75	Above Average
50% to less than 55%	C+	2.50	Average
45% to less than 50%	C	2.25	Below Average
40% to less than 45%	D	2.00	Pass
Less than 40%	F	0.00	Fail

Class Make-up Procedure

Missed class will be taken at a convenient free class hour.

Notional Hour

	Credits	Notional Hour (Guided by BNQF)	Face to Face/ Class Time	STL, Preparatory time etc.
Theory	12	12*40=480 Hours	For 1 theory course (3 credit) 3 Hour/week * 14 week=42 hours for 4 theory courses (3*4 credit) 42*4=168 hours	At least 4 hours
Lab	1	1*60=60 Hours	For 1 lab course (1 credit) 1.5 Hour/week * 14 week=21 hours	
Total	13	540 Hours	189 Hours	

Textbook/Recommended Readings

Ronald J. Tocci, Neal S. Widmer, Gregory L. Moss, Digital Systems: Principles and Applications, 12th Edition, Pearson Education International, 2021, ISBN: 978-0131725799.

Reference Books/ Other Supplementary Readings

Mano, M. Morris. Digital Logic & Computer Design, 4th edition, Prentice Hall, 2004, ISBN: 0-13-306551-3.

Course Materials and Slides

All course materials and slides will be available in DIU Blended Learning Center, and Google classroom. (Everyone is requested to join Google Classroom and BLC account).

Software/Tools used

N/A

Exam Dates

According to the Examination Schedule

Academic Code of Conduct

Academic Integrity:

Academic offenses under the Academic Code of Conduct include plagiarism, personification, physical and online cheating, falsification of a document, and any other dishonest behavior related to gaining academic gain or avoiding evaluation exercises by a student. The university's Disciplinary Committee may decide to impose severe penalties for these offenses.

Special Instructions:

- The tutorial class will be held in accordance with the department's guidelines.

- Attendance at all classes and exams is required of the students. To take the final test, a student needs to have attended at least 70% of classes.
- After ten minutes of the scheduled start time, students will not be permitted to enter the classroom.
- Plagiarism will automatically result in a zero on that exam or assignment.
- There won't often be a make-up exam. However, if a student misses an exam due to a serious sickness, the death of a family member, an emergency involving the family, or humanitarian reasons, they MUST request permission to make up the exam in writing through the course instructor to the chairperson within 48 hours of the exam date. The application must be submitted with the appropriate supporting documentation for the reason(s) for the absence from the exam.
- There won't be a makeup exam for the final exam. However, if a student is unable to attend the final exam due to a serious illness, a family member's death, an emergency, or humanitarian reasons, they MUST request an incomplete grade in writing from the course instructor via the chairperson within 48 hours of the exam date. Along with the application, appropriate supporting documentation for the reason(s) for missing the final exam must be provided. It is the student's duty, in consultation with the course instructor, to schedule an incomplete exam by the deadline specified in the academic calendar.
- It is required that all cell phones be in silent mode during class and test times.
- Exam cheating is not tolerated at all. Examinees will be penalized for cheating if they are found in possession of cheat sheets, used or not; if they write on their palms, the backs of calculators, chairs, or adjacent walls; if they copy from cheat sheets or other sources; if they copy from other examiners, etc. Cheating only carries a single, multi-semester expulsion, as determined by the university's disciplinary committee.

Appendix

Program Learning Outcomes (PLOs)

No.	Program Learning Outcomes
PLO1	Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PLO2	Identify, formulate, research and analyze complex engineering problems and reach substantiated conclusions using the principles of mathematics, the natural sciences and the engineering sciences.
PLO3	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety and of cultural, societal and environmental concerns.
PLO4	Conduct investigations of complex problems, considering experimental design, data analysis and interpretation and information synthesis to provide valid conclusions.
PLO5	Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modeling, to complex engineering activities with an understanding of their limitations.
PLO6	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PLO7	Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development.
PLO8	Apply ethical principles and commit to the professional ethics, responsibilities and the norms of the engineering practice.
PLO9	Function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings.
PLO10	Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.
PLO11	Demonstrate knowledge and understanding of engineering and management principles and apply these to one's work as a team member or a leader to manage projects in multidisciplinary environments.
PLO12	Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.

Learning Domain (Bloom's taxonomy)

- (a) Cognitive Domain (Knowledge):** The cognitive domain aims to develop the mental skills and the acquisition of knowledge of the individual.

Domain Level	Definition
Remember (C1)	Retrieving, recalling, or recognizing information from memory.
Understand(C2)	Changing from one form of representation to another; illustrating a concept; drawing conclusions; determining cause and effect

Apply(C3)	Using learned materials, students can use/apply information in a new way.
Analyze(C4)	Breaking material or concepts into parts, determining how the parts related or interrelated to one another or to an overall structure or purpose.
Evaluate(C5)	Assessing, making judgments and drawing conclusions from ideas, information, or data.
Create(C6)	Developing a hypothesis; devising a procedure; inventing a product

(b) Affective Domain (Attitude): The affective domain includes how learners deal with things emotionally, such as feelings, values, appreciation, enthusiasms, motivations, and attitudes.

Domain Level	Definition
Receiving (A1)	Being aware of or attending to something in the environment.
Responding(A2)	Showing some new behaviors as a result of experience.
Valuing(A3)	Showing some definite involvement or commitment.
Organization(A4)	Integrating a new value into one's general set of values, giving it some ranking among one's general priorities.
Characterization by value(A5)	Acting consistently with the new value.

(c) Psychomotor Domain (skill): Includes physical movement, coordination, and use of the motor-skill areas.

Domain Level	Definition
Imitating (P1)	Attempted copying of a physical behavior
Manipulation(P2)	Reproducing activity from instruction or memory
Precision(P3)	Fine tuning. Making minor adjustments in the physical activity in order to perfect it.
Articulation(P4)	Adapting and integrating expertise to satisfy a non- standard objective
Naturalization(P5)	Automated, unconscious mastery of activity and related skills at strategic level

BNQF Skills (4 year's Bachelors):

The Learning Outcome Domains are Fundamental Domain, Social Domain, Thinking Domain and Personal Domain. These domains are classified below.

Fundamental Skills:

- Demonstrate knowledge and critical understanding of the well-established principles of his/her field of study, and of the way in which those principles have developed.
- Apply underlying concepts and principles outside the context in which they were first studied, including, where appropriate, the application of those principles in an employment context.
- Apply knowledge and skills in addressing issues/solving problems with minimal supervision.
- Evaluate critically the appropriateness of different approaches to solving problems in his/her field of study.
- Support supervision of junior staff via a mentor or a leader/manager.
- Display advanced digital literacy which is adequate to perform complex tasks and bring about solutions.

Social Skills:

- Communicate and interact effectively and clearly, ideas, information, problems and solutions as a team to peers, experts and non-experts in Bangla and English.
- Express her/himself fluently and spontaneously in English and Bangla.
- Use language flexibly and effectively for social, academic and professional purposes.
- Produce clear, well structured, detailed text on complex subjects, showing controlled use of organizational patterns, connectors and cohesive devices in advanced proficiency level of Bangla and English.
- Demonstrate the ability to incorporate entrepreneurial skills in planning daily activities.
- Display advanced civic literacy and knowledge, exercising civic rights and obligations at all levels as well as participating in changes for the improvement of Bangladesh society.

Thinking Skills:

- Exercise very substantial degree of autonomy and often significant responsibility in making
- Judgments / decisions towards the management of self, others and for the allocation of substantial resources.
- Demonstrate professional knowledge and practical skills in both technical and management to lead a team in inexperienced environment.

Personal Skills:

- Engage in self-direction and self-enterprise skills.
- Demonstrate social, professional, environmental and ethical practice/ values.
- Show-case global knowledge and competencies to fulfil employment, entrepreneurial and lifelong learning skills; and contribute significantly to the society.